

Xinhao Qu

CONTACT INFORMATION	900 University Ave., Olmsted Hall 1419 Riverside, CA 92521	<i>E-mail:</i> xqu018@ucr.edu <i>Website:</i> LinkedIn
RESEARCH INTERESTS	Efficient and Reliable LLM Systems, Knowledge Transfer	
EDUCATION	University of California, Riverside , Riverside, CA Ph.D., Applied Statistics, Advisor: Xiaoqian Liu 2024/09 – present Xiamen University , Xiamen, China M.S., Statistics, Advisor: Wei Zhong 2021/09 – 2024/06 Zhengzhou University , Zhengzhou, China B.A., Statistics 2017/09 – 2021/06	
EXPERIENCE	Tencent IEG , Shenzhen, China 2026/06 – 2026/09 Agent Engineering Intern • MCP proxy development for <u>query-level authorization on knowledge base</u> • Trajectory logging and evaluation via AgentLens by Explyt; Self-evolution via SkillOpt-Lite workflow by LMMs-Lab; Deploy runtimes and <u>automate issues on Multica</u> Southern University of Science and Technology , Shenzhen, China 2025/11 – 2026/02 Visiting Graduate Student, Collaborators: Hongxin Wei , Hao Zeng Tencent CSIG , Shenzhen, China 2025/06 – 2025/08 LLM Post-training Intern • <u>Agent node post-training for planning and rendering</u> • AGENTGEN data augmentation; Difficulty-aware knowledge distillation; LoRA SFT; DPO alignment; StructEval renderable evaluation	
PUBLICATIONS	Note: the sign [†] indicates the first authorship; the sign  denotes the corresponding author. [1] X. Qu[†] , Q. Heng, H. Zeng, and X. Liu. An Interpretable and Scalable Framework for Evaluating Large Language Models. <i>Submitted</i> . arXiv:2605.07046v1 . • An interpretable and scalable framework for <u>LLM evaluation via representational learning</u>, with theoretical guarantees for parameter identifiability and algorithmic convergence. • Evaluate on MATH-500 and six benchmarks from the Hugging Face Open LLM Leaderboard v2. Results on model abilities are consistent with established parametric scaling laws, and the recovered item features align with human annotations, <u>capturing fine-grained heterogeneity that average accuracy fails to reflect</u>. [2] H. Zeng, H. Huang, X. Qu , J. Huang, B. Jing, and H. Wei. HyPAC: Cost-Efficient LLMs-Human Hybrid Annotation with PAC Error Guarantees. <i>Submitted</i> . arXiv:2602.02550v1 . • Formulate LLMs-human hybrid labeling as constrained cost optimizations with multiple labeling sources, and provide a provable framework for controlling the labeling error rate. • Provide <u>distribution-free guarantees on both error control and cost optimality</u>: the error rate stays below a user-specified tolerance with high probability, and HyPAC achieves the minimum cost among all threshold-based methods with the same constraint. [3] X. Qu[†]  . Partial Transfer Learning for Causal Estimation Under High-Dimensional Confounding. <i>Major Revision at Journal of Computational and Graphical Statistics</i> . [4] X. Qu[†]  . Automatic Transfer Learning for High-Dimensional Linear Regression. <i>Statistics & Probability Letters</i> , 224, 2025. https://doi.org/10.1016/j.spl.2025.110445 . [5] Y. Hu, X. Qu , and M. Tian. Smoothed GMM for Spatial Quantile Regression Model. <i>Acta Mathematicae Applicatae Sinica</i> , 1-26, 2025. https://doi.org/10.20142/j.cnki.amas.202501018 .	

SOFTWARE	<i>PartialTL</i> : R package for the partial transfer learning method.	
	<i>cBMM</i> : Python implementation of the interpretable and scalable LLM evaluation framework.	
HONORS AND AWARDS	Florence Nightingale David Award for Insightful Statistical Application, UC Riverside	2026
	Morris J. Garber Award in Applied Statistics Endowed Fund, UC Riverside	2026
	Outstanding Graduate Dissertation of Fujian Province	2026
	Dean's Distinguished Award, UC Riverside	2024
	First Class Academic Scholarship, Xiamen University	2021, 2022, 2023
	Graduation Highlight, Zhengzhou University	2021
	First Class Academic Scholarship, Zhengzhou University	2017, 2019, 2020
TEACHING	University of California, Riverside , Riverside, CA	
	<i>Teaching Assistant</i>	
	• STAT 167 (Introduction to Data Science)	Winter 2026
	• STAT 010 (Introduction to Statistics)	Fall 2024, Spring 2026
	Xiamen University , Xiamen, China	
	<i>Teaching Assistant</i>	
	• Introduction to Data Science	Fall 2022
SERVICES	Area Chair in Applications of Machine Learning Session at IAAE Annual Conference (2024)	
	Leader of Machine Learning Group, WISERCLUB, Xiamen University	